TONEVAULT

Guitar Rig Studio

Final Mini-Capstone Presentation · React (Vite/Tailwind) + Laravel REST API

DEVELOPER
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REPOSITORY
github.com/vrsnzz19-glitch/DC4-MINI-CAPSTONE.git

CORE MOTIVATION

The Challenge & Solution Goals

The Challenge

Signal Chain Complexity: Tracking multi-pedal layouts, amp parameters, and custom guitar setups across gigs is error-prone.

Fragmented Data: Absence of structured digital catalogs for gear specifications and reusable tone recipes.

ToneVault Goals

Centralized Catalog: 100+ organized pedals.

Visual Builder: Live signal chain & equalizer output visual.

Rig Management: Complete guitar + amp + pedal recipes.

Role Workflows: Admin review vs user workspace.

SYSTEM BLUEPRINT

Architecture & Role Authorization

Studio Admin

Admin@tonevault.test

Full CRUD over master catalog (117 pedals), categories, and community submission review queue.

Registered Guitarist

guitarist@tonevault.test

Personal workspace to construct custom boards, test interactive signal chains, and publish tone recipes.

Decoupled Tech Stack

Frontend: React SPA + Vite + Tailwind

Backend: Laravel API + Sanctum Auth

Database: Relational MySQL Schema

PRODUCT WALKTHROUGH · PART 1

Guitarist Workspace & Signal Chain

1 Auth & Workspace

Clean auth portal leading directly to saved pedalboard collections like *Ambient Stereo Board* and *Classic Rock Board*.

2 Interactive Visualizer

Stage signal chain view with interactive footswitches and real-time live mix spectrum equalizer meters.

3 Pedal Catalog & Filter

Filter 100+ units by category (Distortion, Delay, Modulation) or search specs like *Ibanez TS9 Tube Screamer*.

PRODUCT WALKTHROUGH · PART 2

Rig Presets & Studio Admin Control

Tone Recipes (Rig Presets)

Links pedalboard builds with guitar models (*Fender Jazzmaster*), tunings (*E Standard*), and Amp EQ controls (*Bass 4, Gain 2, Middle 5, Reverb 3, Treble 6*).

Approved / Pending Status Badges

Studio Admin Dashboard

Admin metrics monitoring 117 total catalog pedals, 8 active pedalboards, 3 verified rig presets, and moderating pending community submissions.

Catalog & Queue Operations

TECHNICAL HANDOVER

Repository, Documentation & Security

Version Control

Hosted on GitHub with organized commit history and modular frontend/backend directories.

Docs Folder (/docs)

Includes setup guides, ERD database models, Postman API collections, and seed commands.

Sanctum Security

Token authentication, server-side request validation, and route authorization middleware.

DELIVERABLES EVALUATION

Delivered Features vs. Proposed Plan

Proposed Feature	Status	Implementation Notes
User Authentication & Roles	✅ **Completed**	Separate Admin & Guitarist views via Laravel Sanctum
Pedal Catalog Management	✅ **Completed**	Live search, category filters, and 110+ pre-seeded items
Interactive Pedalboard Builder	✅ **Completed**	Signal chain visualizer with interactive footswitches
Tone Recipe / Rig Presets	✅ **Completed**	Links guitar model, tuning, amp EQ, and pedalboard builds
Review Submission System	✅ **Completed**	Admin approval workflow for public rig presets

FUTURE ROADMAP

Limitations & Phase 2 Recommendations

Current Scope

- Spectrum analyzer is currently visual simulation.
- Preset sharing is limited to in-app submission views.

Phase 2 Roadmap

- **Web Audio API:** Live audio DSP effect nodes in browser.
- **Community Features:** Ratings, reviews, and preset cloning.
- **Hardware Export:** Export presets to M-Vave or Line 6 formats.